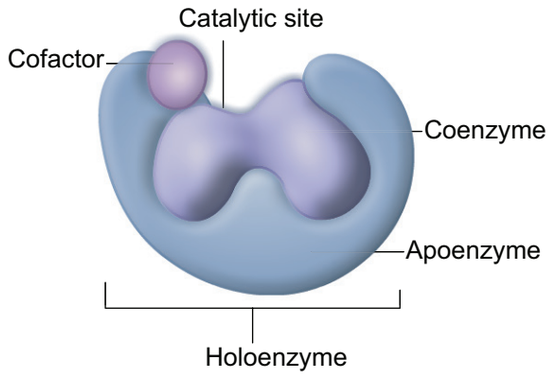


Figure 8.19: Enzyme components

- **Inorganic ions** help to increase the rate of reaction catalysed by enzymes. Example: Salivary amylase activity is increased in the presence of chloride ions.
- **Prosthetic groups** are organic molecules that assist in catalytic function of an enzyme. Flavin adenine dinucleotide (FAD) contains riboflavin (vit B2), the function of which is to accept hydrogen. 'Haem' is an iron-containing prosthetic group with an iron atom at its centre.
- **Coenzymes** are organic compounds which act as cofactors but do not remain attached to the enzyme. The essential chemical components of many coenzymes are vitamins. Eg. NAD, NADP, Coenzyme A, ATP



Ribozyme - Non Protein Enzyme - A Ribozyme, also called as catalytic RNA; is a ribonucleic acid that acts as enzyme. It is found in ribosomes

8.6.4 Classification of Enzymes

Enzymes are classified into six groups based on their mode of action.

Enzymes	Mode of action	General scheme of reaction	Example
Oxidoreductase	Oxidation and reduction (redox) reactions	$A_{red} + B_{ox} \longrightarrow A_{ox} + B_{red}$	Dehydrogenase
Transferase	Transfer a group of atoms from one molecule to another	$A - B + C \longrightarrow A + C - B$	Transaminase, phosphotransfer-ase
Hydrolases	Hydrolysis of substrate by addition of water molecule	$A - B + H_2O \longrightarrow A - H + B - OH$	Digestive enzymes
Isomerase	Control the conversion of one isomer to another by transferring a group of atoms from one molecule to another	$A - B - C \longrightarrow A - C - B$	Isomerase
Lyase	Break chemical bond without addition of water	$A - B \longrightarrow A + B$	Decarboxylase
Ligase	Formation of new chemical bonds using ATP as a source of energy	$A + B + ATP \longrightarrow A - B + ADP + Pi$	DNA ligase

XIth_Botany_Chapter_8_EM.indd 188 05-01-2023 18:27:44